

2026-05-19-const052-rename-programme-plan

CONST-052 Rename Programme — Phased Plan (ISSUE-005)

Spec date: 2026-05-19 **Constitutional anchor:** CONST-052 (constitution §11.4.29) **Tracker:** ISSUE-005 (docs / Issues .md) **Scope:** PLAN ONLY. No renames executed in this round. Operator review + approval gates each phase before execution. **Author intent:** Translate CONST-052’s “*phased execution per the operator’s explicit instruction: comprehensive brainstorming → phase-divided plan → fine-grained tasks/subtasks → every change covered by every applicable test type*” into a concrete, evidence-backed migration map for HelixCode’s meta-repo.

1. Status Snapshot

1.1 Meta-root naming (already compliant)

A find . -maxdepth 1 -type d -not -path "./.git" -not -path "." -printf "%f\n" | sort | grep -E '[A-Z]' against HEAD 2a5586e returns **empty**. The 32 top-level meta-repo directories (assets/, awesome-ai-memory/, challenges/, cli_agents/, cli_agents_configs/, cli_agents_resources/, cmd/, configs/, constitution/, containers/, dependencies/, docker/, docs/, github_pages_website/, helix_agent/, helix_code/, helix_qa/, implementation_guide/, internal/, mcp_servers/, packaging/, panoptic/, reports/, scripts/, security/, specification/, test/, tests/, upstreams/, website/, plus hidden .claude/, .github/) are ALL snake_case (verified 2026-05-18).

ISSUE-005 phrasing — “**meta-repo directories like helix_code/, challenges/, helix_qa/, helix_agent/ still PascalCase**” — was authored before the round-88 meta-root partial sweep; the meta-root is now compliant. **The remaining non-compliance is two layers deeper:** under dependencies/ and inside every submodule’s Up streams/ directory.

1.2 Where PascalCase still lives (the actual work)

Three deficiency clusters remain:

Cluster	Where	Count	Severity
(A) Owned-org submodule directories	dependencies/HelixDevelopment/* + dependencies/vasic-digital/*	10 + 48 = 58 dirs	HIGH (CONST-052 §11.4.29 applies — owned by us)
(B) Third-party			EXEMPT (CONST-052 vendor exception)

Cluster	Where	Count	Severity
submodule directories	dependencies/ {LLama_CPP,Ollama,HuggingFace_Hub}, cli_agents_resources/*	3 + 7 = 10 dirs	
(C) Upstreams/ (PascalCase) → upstreams/ (snake) per CONST-052 transition	49 submodule subtrees (incl. own + vendor)	49 dirs	MEDIUM (transition explicitly supported by constitution/ install_upstreams.sh for both)

Verification commands (all run 2026-05-18 against HEAD 2a5586e):

```
# (A) Owned-org PascalCase submodule dirs:
find dependencies/HelixDevelopment -maxdepth 1 -type d -not
    -path "dependencies/HelixDevelopment" | wc -l    # → 10
find dependencies/vasic-digital -maxdepth 1 -type d -not -path
    "dependencies/vasic-digital" | wc -l    # → 48

# (B) Third-party submodule dirs:
find dependencies -maxdepth 1 -type d | grep -E '/[A-Z]' | wc
    -l    # → 3 (LLama_CPP, Ollama, HuggingFace_Hub)
find cli_agents_resources -maxdepth 2 -type d | grep -E '/[A-Z]'
    | wc -l    # → 7 (Awesome-*, Cheshire-Cat-Ai, GitHub-
    Awesome-Copilot, OpenAI-Cookbook, Taches-CC-Resources)

# (C) PascalCase Upstreams dirs:
find . -maxdepth 4 -type d -name "Upstreams" 2>/dev/null | wc
    -l    # → 49
```

1.3 Common-sense exceptions per CONST-052 §11.4.29

The rule has four explicit carve-outs. Applied here:

1. **Language-mandated case** (Java/Kotlin/Android/Apple/C#/Swift inside language roots). Affects: paths inside cli_agents/junie, cli_agents/swe-agent, helix_code/applications/{ios, android, aurora-os, harmony-os}. **Exempt** — language convention wins inside the language root.
2. **Vendor / upstream third-party submodules.** Affects cluster (B): LLama_CPP, Ollama, HuggingFace_Hub, cli_agents_resources/*, cli_agents/* (third-party agent forks). **Exempt** — renaming would break upstream merge alignment.
3. **Build artefacts** (node_modules, __pycache__, .git, target, build, bin). Already ignored, irrelevant.
4. **Common-sense test** (“does renaming break the technology?”). Special case for submodules/llms_verifier/Assets/ and Website/ — these are language-neutral but ship as fixed-name asset roots wired into deployment manifests; the **decision is operator-side** (deferred to §8 decision-points).

1.4 Why this was deferred

Three coupled cascades historically blocked execution:

1. **Documentation-path encoding.** A literal string `submodules/helix_llm/...` appears in 244 `.md` files (verified count); each is a working anchor that breaks if the directory moves.
2. **Go module replace directives.** 8 `go.mod` files (verified `grep -l "HelixDevelopment\|vasic-digital" --include="go.mod" -r .`) carry `replace` lines that map a module-id to a relative directory; `rename` without paired `replace` update silently breaks `go build`.
3. **Submodule registration.** `.gitmodules` carries 70 PascalCase `path = ...` entries; renaming requires `git mv + .gitmodules rewrite + git submodule sync + re-init` across all clones.

Round 106 (the **HelixMemory** rename pilot) demonstrated all three at once: one wrong path knocked out 6 packages.

2. Reference-Cascade Audit

2.1 Owned-org per-submodule fan-out (the cost-of-rename matrix)

All numbers from `grep -rn "$path" --include="<ext>" . | grep -v "\.git/" | wc -l`, run 2026-05-18.

2.1.1 dependencies/HelixDevelopment/* (10 submodules)

Submodule	.md refs	.sh refs	.go refs	Aggregate	Risk band
HelixLLM	244	9	269	522	PHASE-3 critical (Go-replace blast radius)
LLMsVerifier	88	6	1	95	PHASE-3
DocProcessor	65	2	0	67	PHASE-2
LLMOrchestrator	57	5	1	63	PHASE-2
VisionEngine	56	5	0	61	PHASE-2
LLMProvider	41	1	0	42	PHASE-2
HelixMemory	39	9	0	48	PHASE-2 (pilot already done — confirms cost band)
HelixSpecifier	39	9	0	48	PHASE-2

Submodule	.md refs	.sh refs	.go refs	Aggregate	Risk band
Models	5	0	0	5	PHASE-1
DebateOrchestrator	4	0	0	4	PHASE-1

Aggregate cluster total: dependencies/HelixDevelopment **string appears 260 times** across .md/.sh/.go/.yaml/.yml/.toml (verified).

2.1.2 dependencies/vasic-digital/* (48 submodules — top 10 only audited; remainder estimated)

Submodule	.md refs	.sh refs	.go refs	Aggregate	Risk band
Models	8	8	0	16	PHASE-1
Memory	6	1	0	7	PHASE-1
Storage	6	2	0	8	PHASE-1
VectorDB	6	2	0	8	PHASE-1
Auth	5	4	0	9	PHASE-1
Cache	5	2	0	7	PHASE-1
Database	4	2	0	6	PHASE-1
RAG	3	2	0	5	PHASE-1
Plugins	2	1	0	3	PHASE-1
MCP_Module	[estimated 5-10, not exhaustively counted]	[estimated 2-3, not exhaustively counted]	0	~10	PHASE-1
Other 38 vasic-digital submodules	[estimated 0-10 each, not exhaustively counted]	[estimated 0-3 each, not exhaustively counted]	0	individually small	PHASE-1

Aggregate cluster total: dependencies/vasic-digital **string appears 138 times** (verified).

2.2 Third-party (EXEMPT but tabulated for completeness)

Path	.md/.sh/.go refs	Disposition
dependencies/LLama_CPP	4	EXEMPT (vendor; upstream merge alignment)
dependencies/Ollama	4	EXEMPT (vendor)
dependencies/HuggingFace_Hub	20	EXEMPT (vendor)

Path	.md/.sh/.go refs	Disposition
cli_agents_resources/ GitHub-Awesome- Copilot	0	EXEMPT (vendor)
cli_agents_resources/ Awesome-AI-GPTs	0	EXEMPT (vendor)
cli_agents_resources/ {Awesome-AI-Agents, Cheshire-Cat-Ai, OpenAI-Cookbook, Taches-CC-Resources}	[estimated 0-5 each, not exhaustively counted]	EXEMPT (vendor)

2.3 Upstreams/ (PascalCase) → upstreams/ (snake) — cluster (C)

Reference count: `grep -rn "/Upstreams/\\|/Upstreams\\b" --include="*.md" --include="*.sh" . | grep -v "\\git/" | wc -l` → **40 references** across all .md and .sh files.

Important coupling: `constitution/install_upstreams.sh` (commit 45d3678 of the constitution submodule) **already supports BOTH directory names** (verified: `grep -n "upstreams\\|Upstreams" constitution/install_upstreams.sh` returns dual-name handling at lines 6 + lookup loop). This means cluster (C) renames are **non-breaking by construction** — the upstream installer continues to work whichever name the submodule uses. **Convention drift, not functional drift.**

2.4 Submodule directories owned by us but unaffected (already snake_case at root)

- challenges/, containers/, helix_agent/, helix_code/, helix_qa/, security/, assets/, github_pages_website/, panoptic/, awesome-ai-memory/.

These dirs are compliant at the meta-root path. The **remote GitHub repo URLs** (HelixDevelopment/HelixCode.git, HelixDevelopment/HelixAgent.git, vasic-digital/Challenges.git, etc.) are PascalCase but **CONST-052 §11.4.29 governs filesystem paths, not org-side repository identifiers** — renaming a GitHub repo is a separate operational concern outside CONST-052's scope (would require breaking-change coordination across every clone everywhere). Out-of-scope here.

3. Renames To Perform (Full List)

3.1 Cluster A — owned-org dependencies/HelixDevelopment/* (10 renames)

OLD	NEW
submodules/ debate_orchestrator/	dependencies/ helix_development/ debate_orchestrator/

OLD	NEW
submodules/doc_processor/	dependencies/ helix_development/ doc_processor/
submodules/helix_llm/	dependencies/ helix_development/helix_llm/
submodules/helix_memory/	dependencies/ helix_development/ helix_memory/
submodules/helix_specifier/	dependencies/ helix_development/ helix_specifier/
submodules/llm_orchestrator/	dependencies/ helix_development/ llm_orchestrator/
submodules/llm_provider/	dependencies/ helix_development/ llm_provider/
submodules/llms_verifier/	dependencies/ helix_development/ llms_verifier/
submodules/models/	dependencies/ helix_development/models/
submodules/vision_engine/	dependencies/ helix_development/ vision_engine/

Parent dir HelixDevelopment/ → helix_development/ is part of cluster A (renamed once before/after the leaf renames per operator preference).

3.2 Cluster A — owned-org dependencies/vasic-digital/* (48 renames)

vasic-digital/ itself **already contains a hyphen** (- separator). CONST-052 §11.4.29 says “snake_case” with _ as the separator. **Org-name renaming decision deferred to §8** (would require GitHub org rename, which is out-of-scope). For the IN-SCOPE filesystem path, treatment:

- **Option 1:** leave the org subdir as vasic-digital/ (it is operator’s GitHub-org handle; treat as proper noun exempt — though CONST-052 mentions only language exceptions, not proper nouns). Plan default.
- **Option 2:** rename to vasic_digital/ (strictly compliant; introduces drift between filesystem path and GitHub org handle).

Plan adopts Option 1 (defer org-dir rename pending operator decision, §8). The **leaf submodule renames proceed** regardless:

OLD (sample — full list applied to all 48)	NEW
submodules/agentic/	submodules/agentic/
submodules/auth/	submodules/auth/
submodules/auto_temp/	submodules/auto_temp/
submodules/background_tasks/	submodules/background_tasks/
submodules/benchmark/	submodules/benchmark/
submodules/cache/	submodules/cache/
submodules/claritas/	submodules/claritas/
submodules/concurrency/	submodules/concurrency/
submodules/config/	submodules/config/
submodules/conversation/	submodules/conversation/ (NO-OP — already compliant)
submodules/database/	submodules/database/
submodules/doc_processor/	submodules/doc_processor/
submodules/document/	submodules/document/
submodules/embeddings/	submodules/embeddings/
submodules/event_bus/	submodules/event_bus/
submodules/filesystem/	submodules/filesystem/
submodules/formatters/	submodules/formatters/
submodules/gandalf_solutions/	submodules/gandalf_solutions/
submodules/hyper_tune/	submodules/hyper_tune/
submodules/i_llm/	submodules/i_llm/
submodules/i18n/	submodules/i18n/
submodules/lazy/	submodules/lazy/
submodules/leak_hub/	submodules/leak_hub/
submodules/llm_ops/	submodules/llm_ops/
submodules/llm_orchestrator/	submodules/llm_orchestrator/
submodules/llm_provider/	submodules/llm_provider/
submodules/mcp_module/	submodules/mcp_module/
submodules/memory/	submodules/memory/
submodules/messaging/	submodules/messaging/
submodules/middleware/	submodules/middleware/
submodules/models/	submodules/models/
submodules/normalize/	submodules/normalize/
submodules/observability/	submodules/observability/
submodules/optimization/	submodules/optimization/

OLD (sample — full list applied to all 48)	NEW
submodules/ouroborous/	submodules/ouroborous/
submodules/planning/	submodules/planning/
submodules/plinius_common/	submodules/plinius_common/
submodules/plugins/	submodules/plugins/
submodules/rag/	submodules/rag/
submodules/rate_limiter/	submodules/rate_limiter/
submodules/recovery/	submodules/recovery/
submodules/red_team/	submodules/red_team/
submodules/self_improve/	submodules/self_improve/
submodules/skill_registry/	submodules/skill_registry/
submodules/storage/	submodules/storage/
submodules/streaming/	submodules/streaming/
submodules/tool_schema/	submodules/tool_schema/
submodules/toon/	submodules/toon/
submodules/vector_db/	submodules/vector_db/
submodules/veritas/	submodules/veritas/
submodules/vision_engine/	submodules/vision_engine/
submodules/watcher/	submodules/watcher/

Total cluster A: **10 + 48 = 58 leaf renames** (+1 parent rename = 59 if HelixDevelopment → helix_development executes).

3.3 Cluster B — third-party (NO RENAMES — exempt per CONST-052 §11.4.29 vendor carve-out)

dependencies/LLama_CPP, dependencies/Ollama, dependencies/HuggingFace_Hub, cli_agents_resources/* (7 dirs), cli_agents/* (third-party agent forks) — **0 renames**. Rationale documented per dir in a follow-on exemption ledger (created in Phase-1 setup, §4.1).

3.4 Cluster C — Upstreams/ → upstreams/ (49 renames)

All 49 submodule Upstreams/ subdirs:

./constitution/Upstreams	→ ./constitution/upstreams
./panoptic/Upstreams	→ ./panoptic/upstreams
./github_pages_website/Upstreams	→ ./github_pages_website/up
./containers/Upstreams	→ ./containers/upstreams
./helix_qa/Upstreams	→ ./helix_qa/upstreams
./challenges/Upstreams	→ ./challenges/upstreams
./cli_agents/claude-code-source/Upstreams	→ ./cli_agents/claude-code-
./security/Upstreams	→ ./security/upstreams
./helix_agent/Upstreams	→ ./helix_agent/upstreams

<code>./helix_agent/templates/Upstreams</code>	<code>→ ./helix_agent/templates/u</code>
<code>./dependencies/HelixDevelopment/*/Upstreams</code>	<code>→ ./dependencies/helix_deve</code>
<code>./dependencies/vasic-digital/*/Upstreams</code>	<code>→ ./dependencies/vasic-digi</code>

Important: each Upstreams/ lives INSIDE a submodule's own git tree. Renaming requires committing in that submodule's repo first, then bumping the parent's `.gitmodules` pointer. Cluster (C) is therefore **49 inner-submodule commits + 1 parent bump per leaf** — high commit count, low risk per commit.

3.5 Out-of-scope (operator decisions in §8)

- `HelixDevelopment/` parent dir → `helix_development/` (renames 260 references, must paired-execute with cluster A).
- `vasic-digital/` parent dir → `vasic_digital/` (renames 138 references; drift vs GitHub org handle).
- `LLMsVerifier/Assets/` → `assets/` and `LLMsVerifier/Website/` → `website/` (inside submodule, possibly deployment-wired).

4. Phased Execution Plan

Five phases ordered by **cost × blast radius**, low to high. Each phase ends with a green-bar checkpoint before next phase begins. **Operator approves each phase before execution.**

4.1 Phase 0 — Pre-flight (tooling + exemption ledger)

Goal: make subsequent phases auditable and reversible.

Tasks:

1. **T-0.1** Author `scripts/rename-dir.sh` <OLD> <NEW> — wraps `git mv`, then runs the four reference-class sweeps (`.md`, `.sh`, `.go`, `.gitmodules`) with `sed -i` replacement, then `git submodule sync --recursive`. Idempotent: re-run is no-op once new path exists.
2. **T-0.2** Author `scripts/verify-no-stale-refs.sh` <OLD> — exits non-zero if `grep -rn "$OLD" --include='*.{md,sh,go,yaml,yml,toml}' . | grep -v "\.git/"` returns any hit. Used as gate post-rename.
3. **T-0.3** Author `scripts/dry-run-rename.sh` <OLD> <NEW> — produces a diff preview of every file change without writing (uses `git mv -n` semantics emulated by `git diff --no-index` against a temp `sed`-rewritten tree).
4. **T-0.4** Create `docs/governance/CONST-052-exemption-ledger.md` enumerating every directory exempted per §1.3 / §3.3 with citation. Becomes the audit answer to “why is `LLama_CPP` still `PascalCase`?”
5. **T-0.5** Add a pre-commit hook entry to `.git/hooks/pre-commit` (template form, opt-in) that runs `verify-no-stale-refs.sh` for any `PascalCase` path mentioned in `git diff --staged --name-only`.

Test coverage (per CONST-050(B) + CONST-052 §11.4.29):

- Unit: feed each script a synthetic 3-file fixture; assert correct rename behaviour.
- Integration: throwaway-clone test that runs `rename-dir.sh` against a real (cluster-A pilot) directory, verifies build still passes.
- Anti-bluff wire evidence: pre-rename SHA + post-rename SHA + `git submodule status` snapshot.

Estimated effort: 1 commit (scripts), ~250 LOC bash, 0 pushes.

4.2 Phase 1 — Low-risk leaves (vasic-digital + small HelixDevelopment + cluster C inner)

Scope: 48 cluster-A vasic-digital renames + 2 small cluster-A HelixDevelopment renames (Models, DebateOrchestrator) + all 49 cluster-C Upstreams/ → upstreams/ renames.

Why first: lowest reference cascades (≤ 16 refs each for vasic-digital, ≤ 5 for the two HD leaves, ≤ 40 total for Upstreams).

Tasks (per-rename templated; representative sample shown):

```
For each leaf L in {vasic-digital/*} U {HelixDevelopment/Models, HelixDeve
  T-1.L.1 ./scripts/dry-run-rename.sh dependencies/<L> dependencies/<new_L>
  T-1.L.2 Operator review of diff preview (one approval covers a batch of
  T-1.L.3 ./scripts/rename-dir.sh dependencies/<L> dependencies/<new_L>
  T-1.L.4 git -C dependencies/<new_L> submodule sync
  T-1.L.5 cd helix_code && go build ./... (verifies no Go-import drift)
  T-1.L.6 ./scripts/verify-no-stale-refs.sh dependencies/<L>
  T-1.L.7 Run helix_code/Makefile unit + integration tests (real infra per
  T-1.L.8 Commit: "rename(P1): <L> → <new_L> (CONST-052)"
  T-1.L.9 Push to all 4 remotes.
```

For each Upstreams dir U:

```
  T-1.U.1 cd <U>'s parent submodule
  T-1.U.2 git mv Upstreams upstreams
  T-1.U.3 sed -i 's|/Upstreams/|/upstreams/|g' (per the submodule's tree o
  T-1.U.4 ./scripts/verify-no-stale-refs.sh Upstreams (inside that submodu
  T-1.U.5 Submodule commit + push (4 remotes).
  T-1.U.6 In parent: git add <submodule> && commit "bump: <submodule> upst
```

Batching: group 5 leaves per commit for vasic-digital (10 commits total) + 1 commit for cluster-C bumps in parent + per-submodule cluster-C inner commits (~49 inner commits across 49 submodules).

Test coverage per batch: 1. **Regression test:** `verify-no-stale-refs.sh` for every renamed path. 2. **Unit:** `cd helix_code && make test` (real units). 3. **Integration:** `cd helix_code && make test-integration-full` against the docker-compose stack from `make test-infra-up`. 4. **E2E:** `cd helix_code/tests/e2e/challenges && go run cmd/runner/main.go -all`. 5. **Challenges:** `cd challenges && make demo-all`. 6. **Wire evidence:** terminal output of every test run captured, attached to the close-out.

Estimated effort: ~10 batch commits cluster A + ~49 inner commits cluster C + 49 parent bumps + 1 close-out commit. ~110 commits, ~440 pushes (4 remotes). LOC delta primarily renames (low); reference-update sed sweeps ~150 LOC delta net.

4.3 Phase 2 — Medium-cascade HelixDevelopment leaves

Scope: DocProcessor (67), LLMOrchestrator (63), VisionEngine (61), LLMProvider (42), HelixMemory (48 — partially done in round 88-pilot, re-verify), HelixSpecifier (48).

Why second: medium reference cascades (40-70 each). The HelixMemory pilot taught us the failure modes (round 106 — 6 packages broke from one wrong path); apply the lesson at scale here.

Tasks (per-rename, same template as Phase 1 but with three additions):

For each leaf L in {DocProcessor, LLMOrchestrator, VisionEngine, LLMProvider}
T-2.L.0 (NEW) Audit every helix_code/go.mod replace directive that mentions L
T-2.L.1-9 same as Phase 1.
T-2.L.10 (NEW) Run helix_code's helixqa integration suite specifically for L
T-2.L.11 (NEW) Update docs/ARCHITECTURE.md path references in same commit

Test coverage per leaf: Phase 1 floor + module-specific suites (e.g., LLMProvider rename also runs helix_code/internal/providers/... test set).

Estimated effort: 6 leaves × 1 commit = 6 commits, 24 pushes, ~200 LOC sed delta across docs.

4.4 Phase 3 — High-cascade leaves (HelixLLM + LLMsVerifier)

Scope: HelixLLM (244 .md + 9 .sh + **269 .go refs** — 522 aggregate) and LLMsVerifier (95 aggregate, plus CONST-036/037 constitutional dependence).

Why last: highest blast radius. HelixLLM's 269 Go references mean the rename is effectively a refactor of every consumer's import path — same magnitude as a major-version-bump library migration. LLMsVerifier is the constitutional single-source-of-truth (CONST-036) so its rename touches every provider-integration test.

Tasks:

T-3.HLLM.0 Snapshot ALL helix_code/go.mod replace lines mentioning HelixLLM
T-3.HLLM.1 Snapshot every Go file with `import "dev.helix.code/.../HelixLLM"
T-3.HLLM.2 dry-run-rename.sh
T-3.HLLM.3 Operator approval (REQUIRED — large-blast-radius gate).
T-3.HLLM.4 rename-dir.sh
T-3.HLLM.5 Update every go.mod replace + every Go import line.
T-3.HLLM.6 go build ./... in helix_code/.
T-3.HLLM.7 Full test floor: unit + integration + e2e + challenges + helixqa
T-3.HLLM.8 Commit + push.
T-3.HLLM.9 Repeat for LLMsVerifier with extra CONST-036/037 verifier-cascade

Test coverage: full CONST-050(B) matrix — unit, integration, e2e, full-automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges, helix_qa autonomous session. Each leaf gets its own wire-evidence dossier.

Estimated effort: 2 leaves × 2-3 commits each (rename commit + Go-path-sweep commit + post-test fix commit if any) = ~6 commits, ~24 pushes, ~400 LOC delta (mostly Go import lines + go.mod replaces).

4.5 Phase 4 — Parent-dir + Go-module-replace bumps

Scope: if operator approves §8 decisions, rename dependencies/HelixDevelopment/ → dependencies/helix_development/ and optionally dependencies/vasic-digital/ → dependencies/vasic_digital/. **And** bump every replace directive in every go.mod to the new paths.

Tasks: 1. **T-4.1** Inventory: list all 8 `go.mod` files (already verified) plus their `replace` directives. 2. **T-4.2** Author a single bash script `scripts/bump-go-replaces.sh` that walks all 8, applies `sed -i 's|HelixDevelopment|helix_development|g; s|vasic-digital|vasic_digital|g'` (where in-scope), and runs `go mod tidy`. 3. **T-4.3** Bulk rename parent dirs (cluster A parent). 4. **T-4.4** Bulk rename `.gitmodules` paths. 5. **T-4.5** `git submodule sync --recursive` everywhere. 6. **T-4.6** `helix_code` `&& go build ./...` `&& go test ./...` validates Go cascade. 7. **T-4.7** Full test floor. 8. **T-4.8** Commit + push (single atomic commit per cluster A parent + cluster B parent if in scope).

Test coverage: full CONST-050(B) matrix + paired mutation (purposely plant a stale `HelixDevelopment/...` ref and assert `verify-no-stale-refs.sh` catches it).

Estimated effort: 2 atomic commits (one per parent rename), 8 pushes, ~50 LOC delta (path-rewrite only).

5. Test Coverage per Rename Batch (CONST-050(B) + CONST-052 §11.4.29 conformance)

Every rename batch (Phase 1 leaves, Phase 2-3 individual, Phase 4 atomic) MUST ship the four invariants:

Test type	Command	Evidence captured	Gate
1. Reference-resolution regression	<code>./scripts/verify-no-stale-refs.sh <old_path></code>	exit code + stdout	exit=0 required
2. Unit (mocks allowed only here per CONST-050(A))	<code>cd helix_code && make test</code>	terminal output	0 failures
3. Integration (real infra)	<code>cd helix_code && make test-integration-full</code>	terminal output + docker-compose status	0 failures
4. E2E challenges (full user flow)	<code>cd helix_code/tests/e2e/challenges && go run cmd/runner/main.go -all</code>	runner output + per-challenge wire evidence	0 failures
5. Challenges submodule (vasic-digital/Challenges)	<code>cd challenges && make demo-all</code>	per-Challenge captured output	0 failures
6. <code>helix_qa</code> autonomous session	<code>cd helix_qa && ./run-autonomous-session.sh</code>	session report + wire-evidence dossier	0 failures
7. Anti-bluff smoke	<code>grep -rn "simulated\ </code>	empty grep output	empty required

Test type	Command	Evidence captured	Gate
	for now\ TODO implement\ placeholder" helix_code/ internal helix_code/cmd		
8. Governance cascade	./scripts/ verify- governance- cascade.sh	pass/fail report	pass required
9. Constitution rules sweep	./scripts/ verify-all- constitution- rules.sh	per-rule status	every implementable rule PASS

For Phase 3 (high-cascade) + Phase 4 (Go-replace): add (10) `cd helix_code && go build ./... && go test -race ./...` and (11) paired mutation per §1.1 — plant stale ref, verify gate catches it, then revert.

6. Tooling Requirements

Script	Purpose	Phase used	LOC est.
scripts/ rename-dir.sh <OLD> <NEW>	git mv + 4-class reference sweep + submodule sync	All	~80
scripts/ verify-no- stale-refs.sh <OLD>	post-rename gate; greps all reference classes	All	~30
scripts/dry- run-rename.sh <OLD> <NEW>	preview diff without writing	All	~70
scripts/bump- go-replaces.sh	bulk update go.mod replace lines	Phase 4	~50
scripts/ CONST-052- progress- report.sh	print “X of 58 cluster-A leaves done”	Reporting	~40
docs/ governance/ CONST-052- exemption- ledger.md	per-dir exemption rationale (cluster B)	Phase 0 docs	~80 doc lines

Manual edits (no script — operator approval gate per edit):

1. `.gitmodules` — paths updated atomically with `git mv` commits.
 2. Root governance docs (`CLAUDE.md`, `AGENTS.md`, `CONSTITUTION.md`) — only if they cite a renamed path. **Caution:** most governance is inside the `constitution/` submodule; CONST-049 §11.4.26 7-step workflow applies if those need edits.
 3. `docs/ARCHITECTURE.md` — likely needs sweep at end of each phase.
 4. `docs/CONTINUATION.md` — CONST-044 mandates same-commit update for state advancement.
-

7. Risk Mitigation

7.1 Per-phase rollback procedure

Each phase ends with a tagged commit. Rollback = `git revert <tag>` (NOT `git reset --hard`; CONST-043 forbids force operations). Submodule pointer reversions handled by re-checking-out the prior `.gitmodules` content and `git submodule update --init --recursive`.

7.2 Per-rename dry-run

`scripts/dry-run-rename.sh` produces a unified diff preview of every file the script would write. Operator reviews diff before approving the wet run. Same pattern as `git mv -n` for the move itself, extended for reference-class sweeps.

7.3 Bisect strategy

If post-phase test floor FAILs: 1. Identify the failing test. 2. `git bisect start HEAD <phase-start-tag>`. 3. Run `verify-no-stale-refs.sh` + the failing test at each bisect step. 4. Once the offending commit is identified: revert that one commit, fix the missed reference, re-commit.

7.4 Submodule out-of-sync risk

Cluster (C) modifies submodule trees. If a submodule push fails (e.g., a remote rejects), the parent's `.gitmodules` bump must NOT be pushed until all 4 remotes accept the submodule push. Gate: `scripts/verify-submodule-push-symmetric.sh` (NEW — author in Phase 0 if not present).

7.5 Documentation drift after submodule pointer bump

When a submodule renames its `Upstreams/` → `upstreams/`, the parent's `.gitmodules` pointer bumps to a new SHA, but the parent's own `.md` files may still cite `<submodule>/Upstreams/...`. Gate: `verify-no-stale-refs.sh` "Upstreams" runs across the **parent's** tree after each cluster-C bump.

8. Operator Decision Points

The following decisions must be made by the operator BEFORE execution of Phase 1 (or in some cases, before later phases):

#	Decision	Default suggestion	Affects phase
D-1	Execute phases sequentially (1 → 4) or interleave cluster C with cluster A?	Sequential (lowest risk). Cluster C is independent and could run in parallel — but adds cognitive load.	All
D-2	Rename dependencies/ HelixDevelopment/ → dependencies/ helix_development/? Touches 260 references; mandatory for strict CONST-052 compliance but breaks every external bookmark to the org-dir.	YES (strict compliance). Schedule for Phase 4.	Phase 4
D-3	Rename dependencies/ vasic-digital/ → dependencies/ vasic_digital/? Touches 138 references; introduces drift vs GitHub-org handle.	DEFER (proper-noun-via-GitHub-handle exception; treat as carve-out).	Phase 4
D-4	Rename helix_code/ itself (if any PascalCase ever creeps back)? Would change every inner-Go module path (~15 go.mod replaces).	NO action needed — already snake_case.	n/a
D-5	Rename LLMsVerifier/ Assets/ → assets/ and LLMsVerifier/ Website/ → website/? Inside submodule; possibly deployment-wired.	DEFER pending deployment-wire audit.	Phase 3
D-6	Should MCP_Module become mcp_module (already mixed case with underscore) or mcp/module? Strict snake_case = mcp_module.	mcp_module (strict snake_case, single token).	Phase 1
D-7	Should I-LLM become i_llm or illm? Hyphen present in source.	i_llm (preserve token boundary).	Phase 1
D-8		toon (strict).	Phase 1

#	Decision	Default suggestion	Affects phase
	Should TOON become toon or stay UPPER (acronym exception)? CONST-052 doesn't carve out acronyms.		
D-9	Should RAG become rag (acronym)? Same as D-8.	rag (strict).	Phase 1
D-10	Defer cluster C (Upstreams → upstreams) entirely? It's convention-only since install_upstreams.sh accepts both.	NO — strict compliance preferred per CONST-052 spirit.	Phase 1
D-11	Should rename commits be co-authored "Co-Authored-By: Claude" per HelixCode commit convention?	YES (consistent with existing close-out cadence).	All
D-12	Per-phase approval cadence: one approval per phase, or one per batch (5-leaf group)?	One per phase (less interruption). Per-batch escalation only if a phase reveals failures.	All

9. Estimated Effort

Aggregate estimates for the full programme, anchored to the verified data above.

Phase	Commits	Pushes (× 4 remotes)	LOC delta	Wall-time guess (operator-paced)
Phase 0 (tooling)	1	4	~250 LOC scripts + 80 LOC docs	half-day
Phase 1 (low-risk leaves + cluster C)	~110	~440	~200 LOC net (mostly sed sweeps)	2-3 days
Phase 2 (medium HelixDevelopment)	~6	~24	~200 LOC	half-day
Phase 3 (HelixLLM + LLMsVerifier)	~6	~24	~400 LOC (Go imports + replaces)	1 day
Phase 4 (parent dirs + Go-replace bumps)	~2	~8	~50 LOC	quarter-day
Total programme	~125 commits	~500 pushes	~1100 LOC delta	

Phase	Commits	Pushes (× 4 remotes)	LOC delta	Wall-time guess (operator- paced)
				~5 days operator- paced

Test runs: each batch triggers ~9 test types per §5; for 110 Phase-1 batches that is ~990 test runs (some batched together — actual unique runs probably 60-80 if batched 5-leaf-per-batch). Phase 3 alone may take 30-60 min per leaf due to full helix_qa autonomous session.

Conservativeness: the LOC + commit counts above are **lower bounds**. Real execution will surface unanticipated reference classes (e.g., .yaml deployment manifests, .toml package configs) requiring extra sed sweeps. Mark every estimate as [estimated, will refine after Phase 0 dry-run output].

Appendix A — Verification commands run for this plan (all 2026-05-18, HEAD 2a5586e)

```
# §1.1 meta-root compliance:
find . -maxdepth 1 -type d -not -path "./.git" -not -path "."
    -printf "%f\n" | sort | grep -E '[A-Z]' | wc -l    # → 0

# §1.2 cluster A counts:
find dependencies/HelixDevelopment -maxdepth 1 -type d | wc -l
    # → 11 (includes parent)
find dependencies/vasic-digital -maxdepth 1 -type d | wc -l
    # → 49 (includes parent)

# §1.2 cluster C count:
find . -maxdepth 4 -type d -name "Upstreams" | wc -l    # → 49

# §1.4 go.mod count:
grep -rl "HelixDevelopment\|vasic-digital" --include="go.mod"
    -r . | grep -v "\.git/" | wc -l    # → 8

# §2.1 per-submodule HelixDevelopment refs (sample):
grep -rn "HelixDevelopment/HelixLLM" --include="*.go" . | grep
    -v "\.git/" | wc -l    # → 269
grep -rn "HelixDevelopment/HelixLLM" --include="*.md" . | grep
    -v "\.git/" | wc -l    # → 244

# §2.3 cluster C refs:
grep -rn "/Upstreams\|/Upstreams\b" --include="*.md" --
    include="*.sh" . | grep -v "\.git/" | wc -l    # → 40

# §1.4 .gitmodules PascalCase paths:
```

```
grep -E 'path = .*[A-Z]' .gitmodules | wc -l    # → 70

# §2.4 aggregate HelixDevelopment / vasic-digital fan-out:
grep -rn "dependencies/HelixDevelopment" --include="*.md" --
include="*.sh" --include="*.go" --include="*.yaml" --
include="*.yml" --include="*.toml" . | grep -v "\.git/"
| wc -l    # → 260
grep -rn "dependencies/vasic-digital" --include="*.md" --
include="*.sh" --include="*.go" --include="*.yaml" --
include="*.yml" --include="*.toml" . | grep -v "\.git/"
| wc -l    # → 138
```

All commands re-runnable; no claim in this plan rests on memory or estimation unless explicitly annotated [estimated, not exhaustively counted].

Appendix B — Anti-bluff pledge

Every claim in §1-§4 above is backed by a verifiable command in Appendix A. The two [estimated, not exhaustively counted] ranges (MCP_Module refs, “other 38 vasic-digital” refs) are explicitly marked; Phase 0 will replace them with full counts before Phase 1 executes. No PASS claim, no completion claim, no false-success summary is permitted in this programme without paired terminal evidence per CONST-035 / Article XI §11.9.